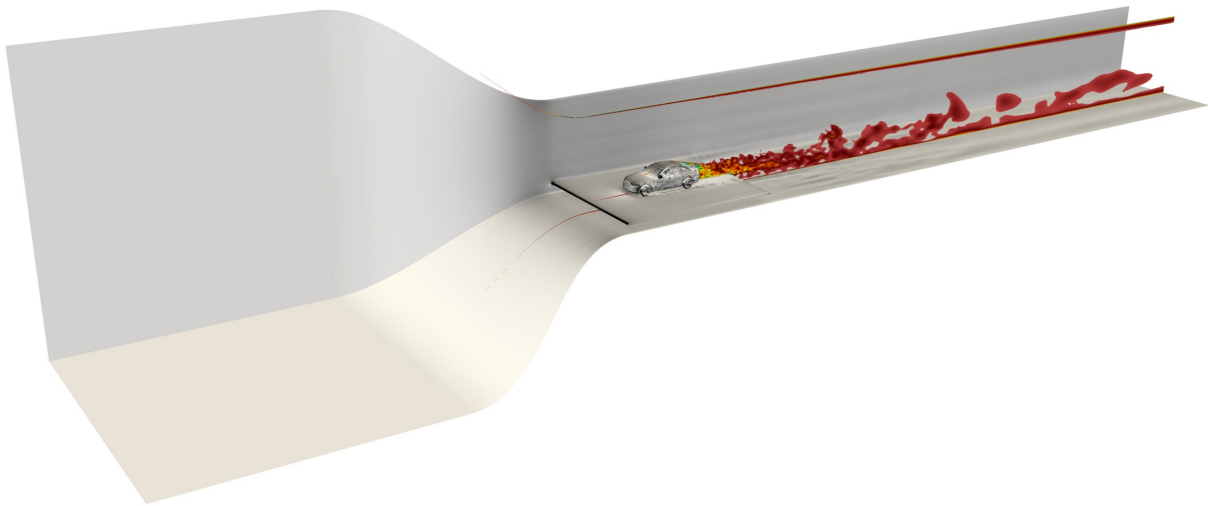


Drag reduction through a sudden pipe contraction

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Introduction

Fluid transport can be considered an essential aspect of numerous engineering applications, from conventional equipment in industry to propulsion systems in rocketry. The formation of such complex systems often necessitates the joining of tubes of different diameters. When such fluids are forced to pass through sudden constrictions, the fluid flow pattern changes, leading to a pressure drop. Although such losses are small in individual cases, their cumulative effect in such extensive fluid transport systems can significantly reduce the efficiency of the system.

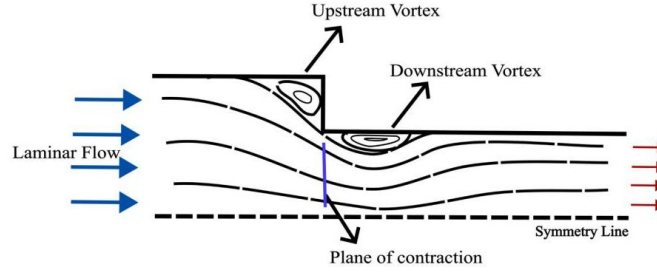


Figure 1: Schematic representation of laminar flow through a sudden pipe contraction, illustrating streamline deflection and the formation of upstream, downstream recirculation zones, and the plane of contraction

Pressure loss in sudden pipe contractions (Fig. 1) stems from two distinct recirculating regions formed upstream and downstream of the contraction plane. The downstream vortex induces a vena contracta effect by reducing the effective cross-sectional area. While this phenomenon was studied extensively by Durst and Loy [1], various mitigation techniques have since been investigated, including the placement of an upstream wall-mounted obstacle as proposed in previous studies [2, 3]. This project utilizes ANSYS CFX to investigate these pressure reduction techniques by simulating the experimental parameters established by [1].

Method

Geometry and Boundary Conditions

The steady-state laminar flow equation was subject to the specified condition. A fully developed velocity profile was applied at the inlet to match the experimental conditions, and the boundary condition at the exit was set to zero static pressure. A no-slip boundary was applied at the walls.

The model is simulated for the range of Reynolds numbers investigated by Durst and Loy, i.e., $398 \leq Re \leq 1213$, to match dynamic similarity against experimental data. In the Durst and Loy experiment, heating oil and Palatinol were used as fluid for matching the refractive index of the test section to the Laser-Doppler Anemometry system, which is irrelevant to our numerical investigation, standard water at 25°C properties are used for all simulation.

Governing Equation & Numerical Solution

The flow is governed by the 3D, steady-state Navier-Stokes equation; continuity and momentum equations were only solved, as the temperature field is irrelevant to the current scope of investigation. The velocity profile is defined at the inlet using an expression with mean velocity and Reynolds number for different values.

ANSYS CFX solver is used, with Residuals(RMS) Converged upto 10^{-6} for all the simulations. Parametrized set is used for simulating different varying parameters such Reynolds Number and Geometric values.

Flow Metrics

A non-dimensional parameter (A_c) is used to quantify the vena contracta using an expression. To quantify the pressure loss, two planes were chosen at the distances of $1.3D$ and D in the upstream and downstream of the plane of contraction. The pressure loss coefficient, C_p , is evaluated using the following expression:

$$C_p = \frac{\Delta p}{\frac{1}{2}\rho U_m^2} = \frac{p_1 - p_2}{\frac{1}{2}\rho U_m^2}$$

where Δp is the static pressure difference between the upstream flow (p_1) and the fully recovered downstream flow (p_2). The denominator represents the dynamic pressure, where ρ is the fluid density and U_m is the mean axial velocity.

Model Verification

The computational mesh was generated using a tetrahedron patch-conforming method, with inflation layers applied near the walls to resolve the boundary-layer effects and recirculating flow regions. A body of influence was also used to provide local mesh refinement near the contraction plane, where strong flow gradients, recirculation, and vena contracta formation occur. The element sizes and local refinement parameters were varied to generate three systematically refined meshes: fine, medium, and coarse. The mesh verification study was performed at $Re_D = 1213$, corresponding to the highest Reynolds number investigated, thereby providing a conservative assessment of the mesh requirements for the present simulations.

The refinement ratios were $r_{21} = 1.502$ and $r_{32} = 1.333$, satisfying the recommended refinement ratio of approximately 1.3 or greater. The vena contracta area, A_c , and pressure-loss coefficient, ΔC_p , were selected as the quantities of interest because they directly characterize the contraction-induced flow behaviour and the associated pressure losses.

The apparent order of convergence, p , was calculated using the iterative formulation given by [4]. Richardson extrapolation was subsequently used to estimate the solution in the limit of zero grid spacing, while the approximate relative error, extrapolated relative error, and fine-grid Grid Convergence Index (GCI) were calculated to quantify the mesh-related discretization uncertainty. For A_c , the calculated apparent order was $p = 3.0485$, with a fine-grid GCI of 0.883%. For ΔC_p , the corresponding values were $p = 2.8114$ and $GCI_{fine}^{21} = 1.8621\%$.

Table 1: Grid Convergence Study and the mesh-related discretization errors

Parameters	$\phi = A_c$	$\phi = \Delta C_p$
N_1, N_2, N_3	498465, 155773, 66973	498465, 155773, 66973
r_{21}	1.502	1.502
r_{32}	1.333	1.333
ϕ_1	2.102	17.109
ϕ_2	2.136	17.654
ϕ_3	2.128	17.755
p	3.05	2.811
ϕ_{ext}^{21}	2.087	16.854
e_a^{21}	1.618%	3.185%
e_{ext}^{21}	0.712%	1.512%
GCI_{fine}^{21}	0.883%	1.862%

The estimated fine-grid discretization uncertainty is therefore below 1% for A_c and below 2% for ΔC_p . These values indicate that the effect of further mesh refinement on the quantities of interest is relatively small compared with the changes observed in the subsequent parametric study. The fine mesh was therefore selected for all remaining simulations as it provides a sufficiently low estimated discretization uncertainty while maintaining a reasonable computational cost.

Results

Model Validation

The numerical model was validated against the experimental data reported by [1]. The height of the vortex region obtained from the experimental data was used to determine the corresponding vena contracta area, A_c . The CFD predictions were compared with the experimental values over the investigated Reynolds-number range of $Re_D = 398$ –1213. The estimated fine-grid discretization uncertainty obtained from the mesh verification study was also considered when assessing the numerical results.

The CFD results reproduce the overall experimental trend, with the vena contracta area decreasing as the Reynolds number increases. The numerical predictions show good agreement with the experimental data over most of the investigated range. At the highest Reynolds number, $Re_D = 1213$, a larger deviation is observed. This difference may be associated with the increasing influence of inertial and transitional flow effects in the smaller pipe section, while the present numerical model assumes laminar flow.

It should also be noted that the experimental determination of A_c is subject to measurement uncertainty. Durst and Loy [1] employed Laser-Doppler Anemometry (LDA) with an effective measuring volume of approximately 200 μm in diameter and 1600 μm in length. Although the relatively small transverse dimension provides good local velocity resolution, the finite measuring volume introduces spatial averaging in regions with strong velocity gradients, particularly close to the boundary of the recirculation region. Since the experimental A_c values are derived from the extent of this recirculation region, the finite spatial resolution limits the precision with which

the vena-contracta boundary can be identified. Consequently, the experimental values should not be regarded as exact reference values, and part of the discrepancy between the CFD and experimental results may arise from experimental measurement uncertainty in addition to numerical discretization and modelling assumptions.

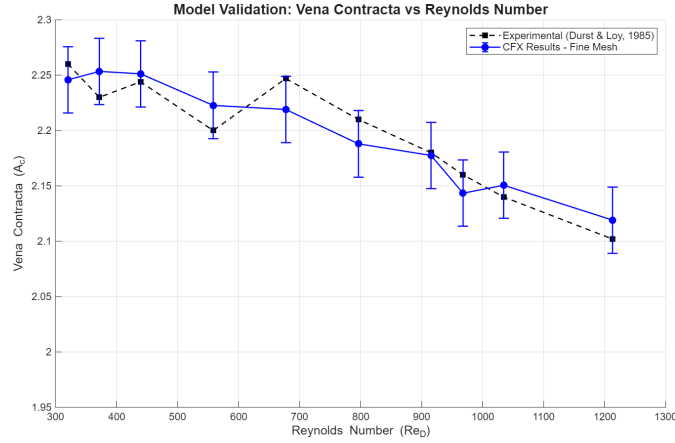


Figure 2: Comparison of the predicted vena contracta area with the experimental data of Durst and Loy [1] over the investigated Reynolds-number range.

Baseline Flow

The baseline flow field, corresponding to the configuration without an obstacle, was investigated at $Re_D = 1213$. Figure 3 shows the velocity field and streamlines in the contraction region. The flow accelerates through the contraction and separates downstream of the contraction, producing recirculating flow regions. The streamlines provide a clearer representation of the separated and recirculating regions than the velocity contour alone and therefore provide an indication of the flow structures responsible for the vena contracta formation.

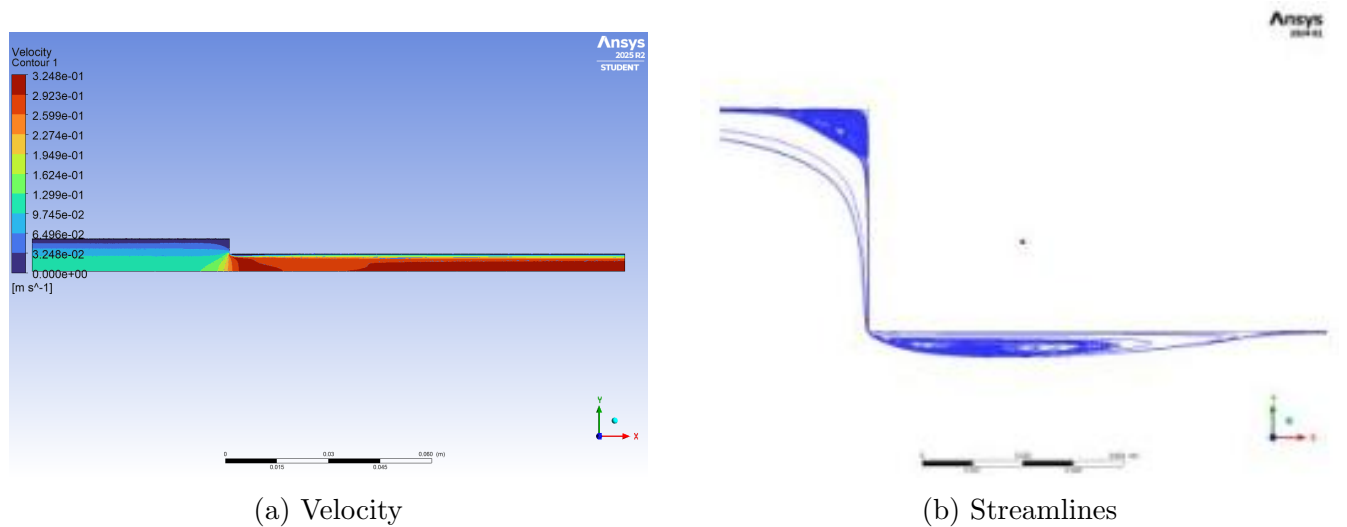


Figure 3: Baseline flow field without an obstacle at $Re_D = 1213$.

The vena contracta and associated recirculation region contribute to the pressure loss through

the contraction. As the Reynolds number increases, the vena contracta area decreases while the pressure loss increases. This behaviour is associated with the increasing influence of inertial effects and flow separation. The increasing deviation between the laminar CFD predictions and the experimental data at the upper end of the Reynolds-number range also indicates the limitations of the laminar-flow assumption for representing the experimental flow under these conditions.

Parametric Study

The obstacle geometry and its upstream position were varied to investigate their influence on the pressure loss and identify a configuration that minimizes ΔC_p . The obstacle height H , width W , and distance L from the contraction plane were varied using the Design of Experiments (DoE) feature in ANSYS. The investigated design space was defined by an obstacle width of $0.5 \text{ mm} < W < 3 \text{ mm}$, an obstacle height of $1 \text{ mm} < H < 5 \text{ mm}$, and an upstream distance from the contraction plane of $2 \text{ mm} < L < 9 \text{ mm}$.

The investigated range of L was selected to cover obstacle positions both close to the contraction and sufficiently upstream to allow the obstacle-induced wake to develop before interacting with the contraction flow. Positions that are too close to the contraction may cause excessive interaction between the obstacle-induced and contraction-induced recirculation regions, whereas positions that are too far upstream allow the wake to dissipate before significantly influencing the contraction flow. The ranges of H and W were similarly selected to provide sufficient blockage to modify the separated flow while avoiding excessive additional pressure loss caused by the obstacle itself.

The pressure loss was found to be most sensitive to the obstacle position L , followed by the obstacle height H and width W . The response surfaces in Fig. 7 illustrate the nonlinear dependence of ΔC_p on the investigated geometric parameters.

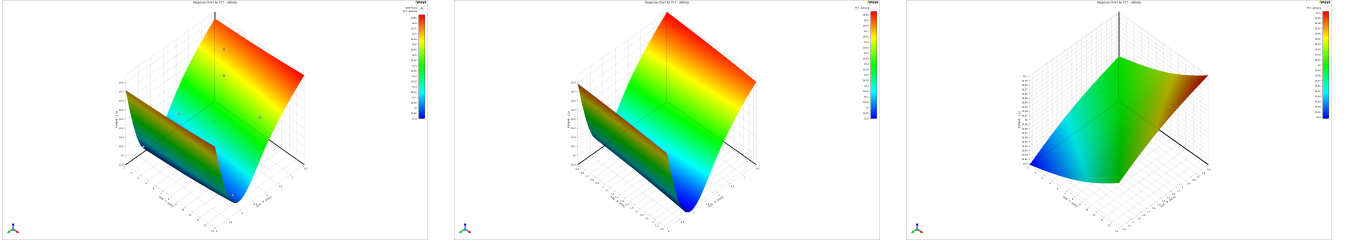


Figure 4: L and H versus ΔC_p . Figure 5: W and H versus ΔC_p . Figure 6: L and W versus ΔC_p .

Figure 7: Response surfaces showing the influence of obstacle geometry and position on the pressure-loss coefficient.

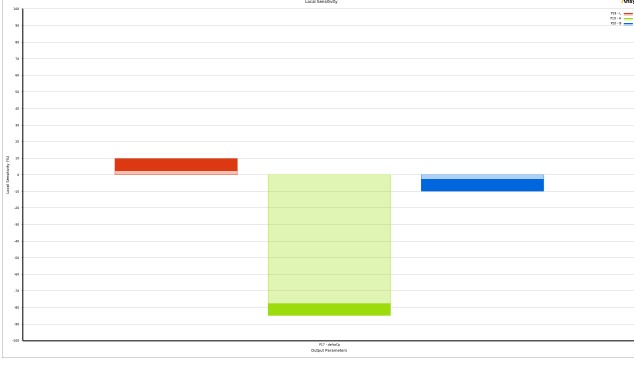


Figure 8: Local sensitivity of ΔC_p to the investigated parameters.

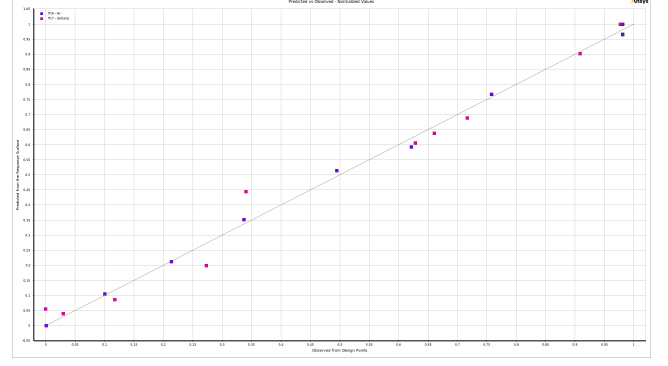


Figure 9: Predicted versus observed response for the DoE model.

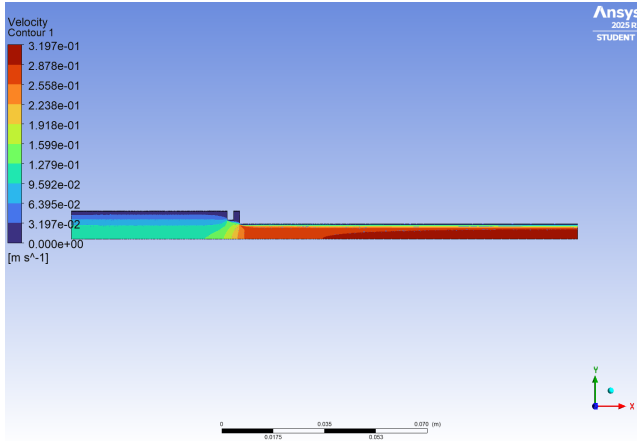
Figure 10: DoE sensitivity and model fitness results.

The response-surface optimization identified an obstacle height of $H = 2.29$ mm. The variation in obstacle position showed a nonlinear influence on the pressure loss, with the effect extending beyond approximately 7 mm. An obstacle position close to the contraction was therefore selected to maximize interaction with the separated flow while avoiding excessive interference between the obstacle-induced and contraction-induced recirculation regions. The minimum pressure loss was obtained at $L = 5$ mm, which was therefore selected for the optimized configuration. The influence of obstacle width at this position was comparatively small, and $W = 0.89$ mm was selected based on the optimization results.

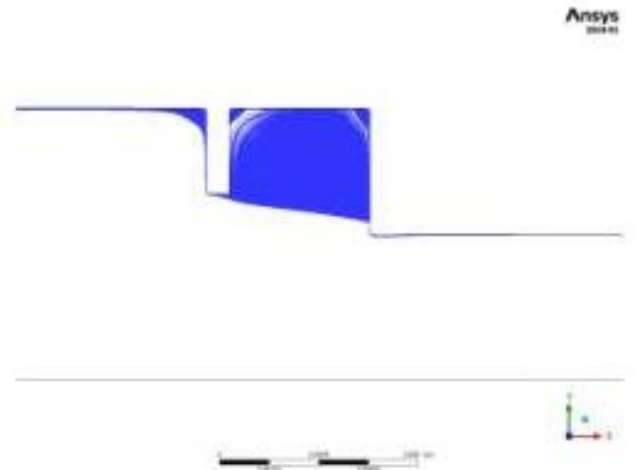
Flow Mechanism of Drag Reduction

To investigate the physical effect of introducing an obstacle upstream of the contraction, a representative non-optimized obstacle configuration was first examined. The obstacle had dimensions of $3.0567 \text{ mm} \times 2.2476 \text{ mm}$ and was positioned at a distance of $L = 2.0457 \text{ mm}$ upstream of the contraction plane. This configuration was used to examine how the presence of an obstacle modifies the flow structure and is distinct from the optimized configuration obtained from the subsequent parametric study.

The velocity field and streamlines in Fig. 11 show that the presence of the obstacle modifies the upstream flow and promotes flow separation and subsequent reattachment before the contraction. The resulting modification of the flow approaching the contraction influences the downstream recirculation region and the development of the vena contracta. The streamline plot provides a clearer representation of these vortical structures than the velocity contour alone.



(a) Velocity



(b) Streamlines

Figure 11: Flow field for a representative non-optimized obstacle configuration with dimensions $3.0567 \text{ mm} \times 2.2476 \text{ mm}$, positioned $L = 2.0457 \text{ mm}$ upstream of the contraction plane at $Re_D = 1213$.

Drag Reduction Performance

The optimized obstacle configuration provides a measurable reduction in pressure loss compared with the baseline configuration. At $Re_D = 398$, the pressure loss is reduced by approximately 2.25%. As the Reynolds number increases, the reduction becomes more pronounced, reaching a maximum reduction of 6.64% at $Re_D = 1213$.

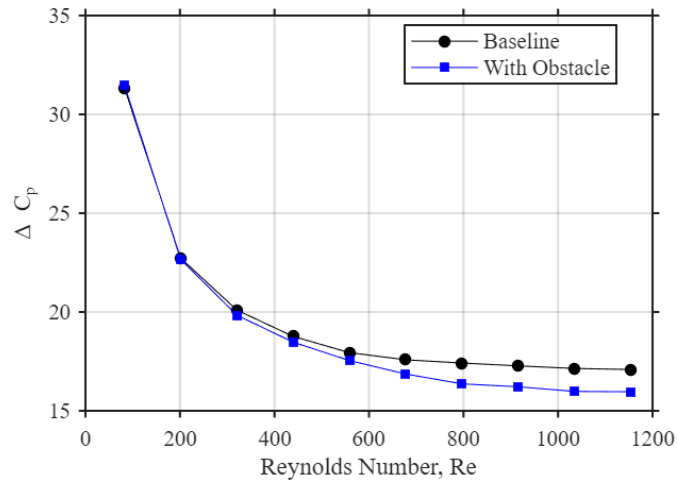


Figure 12: Comparison of the pressure-loss coefficient, ΔC_p , for the optimized obstacle configuration ($H = 2.29 \text{ mm}$, $W = 0.89 \text{ mm}$, $L = 5 \text{ mm}$) and the baseline configuration without an obstacle over the Reynolds-number range $Re_D = 398$ – 1213 .

Discussion

The three-grid mesh verification study demonstrated a low estimated mesh-related discretization uncertainty for both quantities of interest. The fine-grid Grid Convergence Index was 0.883% for the vena contracta area, A_c , and 1.8621% for the pressure-loss coefficient, ΔC_p . The optimized obstacle produced a pressure-loss reduction ranging from approximately 2.25% at $Re_D = 398$ to 6.64% at $Re_D = 1213$. The maximum reduction is approximately 3.6 times the estimated GCI for ΔC_p , indicating that this improvement is appreciably larger than the estimated mesh-related discretization uncertainty and is therefore unlikely to be solely a numerical effect. In contrast, the 2.25% reduction at the lower Reynolds number is only slightly larger than the estimated GCI and should therefore be interpreted with greater caution. Overall, confidence in the predicted drag-reduction trend is strongest for the larger improvements obtained towards the upper end of the investigated Reynolds-number range.

The CFD model was validated against the experimental results of [1] using the vena contracta area as the quantity of interest. The numerical results reproduce the overall decrease in A_c with increasing Reynolds number, although deviations between the numerical and experimental results become more pronounced towards the upper end of the investigated Reynolds-number range. The difference may be associated with the increasing influence of transitional flow effects in the experimental configuration, whereas the present numerical simulations assume laminar flow. In addition, the experimental determination of the vena contracta is subject to measurement uncertainty associated with the finite spatial resolution of the LDA measurements. Consequently, the difference between the numerical and experimental results should be interpreted in the context of both numerical modelling assumptions and experimental uncertainty.

The flow through the sudden contraction is characterized by separation at the sharp contraction edge and the subsequent development of a downstream recirculation region. This separated region reduces the effective cross-sectional area available to the primary flow and contributes to the formation of the vena contracta. The associated pressure and velocity distributions, together with the streamline patterns, demonstrate the strong flow separation and recirculating structures generated by the sudden change in geometry. These structures contribute to the irreversible pressure loss across the contraction.

Introducing a wall-mounted obstacle upstream of the contraction modifies the flow approaching the contraction. The representative non-optimized obstacle configuration investigated in this study demonstrates that the obstacle introduces an additional separation and recirculation region upstream of the contraction. This changes the velocity distribution and streamline pattern before the flow reaches the contraction and consequently modifies the downstream recirculation and vena contracta formation. Rather than eliminating flow separation, the obstacle redistributes the separation and recirculation structures in a manner that can reduce the overall pressure loss.

The parameterized study demonstrated a nonlinear relationship between the obstacle position, L , and the resulting pressure-loss coefficient, ΔC_p . The sensitivity analysis indicated that the obstacle position had the strongest influence on the pressure loss, followed by the obstacle height and width. This behaviour can be attributed to the interaction between the obstacle-induced recirculation region and the contraction-induced separated flow. A suitable obstacle position allows the upstream flow structure to interact beneficially with the contraction flow, whereas an unsuitable position can increase flow disturbance and pressure loss. Therefore, the optimum obstacle configuration is dependent on the operating conditions and the Reynolds-number range considered.

The response-surface optimization identified the configuration $L = 5$ mm, $H = 2.29$ mm, and $W = 0.89$ mm as the configuration providing the lowest pressure loss within the investigated parameter range. The optimized obstacle reduced the pressure loss by approximately 2.25% at $Re_D = 398$, with the reduction increasing to 6.64% at $Re_D = 1213$. The increasing benefit at higher Reynolds numbers indicates that the obstacle-induced modification of the separated flow becomes increasingly effective as inertial effects become more significant within the investigated range.

The maximum reduction of 6.64% is lower than the approximately 16% reduction reported by [2]. Although the contraction ratios used in the two investigations are similar, the operating conditions are different. The present investigation is restricted to the laminar Reynolds-number range up to $Re_D = 1213$, whereas the experiments of [2] were performed at a substantially higher Reynolds number of approximately 5000. The difference in the observed drag-reduction magnitude can therefore be attributed to differences in Reynolds number and the associated flow structures and relative contributions of the different sources of pressure loss. The obstacle primarily modifies the separation and form-drag contribution associated with the contraction, so its relative influence on the total pressure loss is expected to depend on the flow regime and operating Reynolds number.

Conclusions

The present study investigated laminar flow through a sudden pipe contraction using CFD, with particular emphasis on the vena contracta, pressure loss, and passive drag reduction using a wall-mounted obstacle.

A three-grid mesh verification study was performed at the highest investigated Reynolds number, $Re_D = 1213$, following [4]. The resulting fine-grid GCI values were 0.883% for the vena contracta area, A_c , and 1.8621% for the pressure-loss coefficient, ΔC_p . These low estimated discretization uncertainties support the use of the fine mesh for the subsequent simulations.

The numerical model was validated against the experimental data of [1]. The CFD results reproduced the overall trend of decreasing vena contracta area with increasing Reynolds number. Some deviation was observed towards the upper end of the Reynolds-number range, which may be associated with transitional effects not represented by the present laminar numerical model, as well as uncertainty in the experimental determination of the vena contracta.

A passive drag-reduction strategy was subsequently investigated by introducing a wall-mounted obstacle upstream of the contraction. The parameterized study demonstrated that the obstacle position had the strongest influence on the pressure-loss coefficient, followed by the obstacle height and width. Response-surface optimization identified $L = 5$ mm, $H = 2.29$ mm, and $W = 0.89$ mm as the optimum configuration within the investigated parameter range.

The optimized configuration reduced the pressure loss by approximately 2.25% at $Re_D = 398$ and by a maximum of 6.64% at $Re_D = 1213$. The maximum reduction is substantially larger than the estimated fine-grid GCI of 1.8621% for ΔC_p , indicating that the observed improvement is greater than the estimated mesh-related discretization uncertainty. Compared with the larger drag reduction reported by [2], the more modest improvement obtained in the present study is consistent with the lower Reynolds-number and laminar operating regime investigated.

Author Contributions and Declarations

Contribution statement

Author's Name: Mukundan Subbaiya (MS), Neha Mandapaka (NM)

The authors confirm contribution to the report as follows: study conception and design: MS. Author, NM. Author; Setup and simulations: MS. Author, NM. Author; Analysis and interpretation of results: MS. Author, NM. Author; draft report preparation: MS. Author. NM. Author. All authors reviewed the results, approved the final version of the report and declare that each contribution to the work from other peoples published works or unpublished sources have been acknowledged and source of information have been referenced.

Use of generative AI and AI-assisted technologies

During the preparation of this work the author(s) used Gemini in order to fix grammatical errors in the text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the report content.

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